Rafael Petrosian

Los Angeles, CA | 818-588-7056| rafael1@berkeley.edu|website

EDUCATION

Bachelor of Science in Mechanical Engineering | University of California, Berkeley December 2025

TECHNICAL SKILLS

- Co-designed a MEMS device adhering to the MUMPs process, creating a voltage controlled microfluidic thermoactuator exhaust valve incorporating electrostatic latching to prevent fluid leakage while sealed.
- Designed, manufactured, and executed an experiment with a team to evaluate the efficiency of three leading thermal pastes through a controlled heat transfer setup, employing thermocouples and an ESP32 for data acquisition, with data processing and calibration performed in Matlab.
- Conducted rigorous statistical analysis with a team on a dataset of 500 entries, analyzing 12 demographic variables, including sleep patterns, caffeine intake, and exercise habits, using Python and libraries such as Pandas to preprocess data and implement models like neural networks, random forests, and linear regression to identify variable correlations.
- Co-designed and built an automated blind opener powered by a solar-powered ESP32, utilizing sunlight detection for autonomous operation. Python code for the ESP32 was developed and flashed using Thonny to enable autonomous sensing and control.
- Developed an aerodynamically efficient wing model for the Aero SAE plane using Solidworks, enhancing overall flight performance.
- Designed and 3D printed a wheel hub for an advanced rollator as part of a project prototype, utilizing Fusion 360 for the design process.
- Conducted moment of inertia calculations using Excel to determine optimal spar sizing, contributing to the enhancement of structural integrity from McMaster listings.
- Developed a solution for a regenerative cooling circuit within a rocket engine, leveraging Python and LaTeX for accurate modeling and documentation.
- Designed a word search puzzle applet using C++, showcasing programming skills and creative problem-solving.

EXPERIENCE

EDN Aviation | Assistant Engineer Jun 2025 – Aug 2025

- Interpreted mechanical plans and performed measurements to develop precise digital twins in CAD, later used for CNC manufacturing.
- Prepared Bills of Materials (BOMs) for components to support quotation and procurement processes.
- Led an electrical design initiative under the guidance of Ashot Arakelyan, Senior Engineer, by transcribing an electrical plan into CircuitJS, drafting detailed schematics, and developing a test box for a specialized component; additionally created the BOM for its fabrication.
- Automated a packaging and labeling workflow by writing a script for the generation of unique QR codes.

MotoStudent of California | Rear-Suspension Lead Sep 2024 – Dec 2025

- Collaborating with Berkeley students to design, simulate, manufacture, and tune a motorcycle for the 2025 MotoStudent competition in Spain.
- Leading the rear suspension and swing arm design, including components such as the coil, swing arm, rocker, linkage, and other key parts.
- Developing CAD models for rear geometry, exploring solid body and trellis swing arm designs, and conducting preliminary simulations using Onshape.
- Utilizing MotoChassis software for kinematic analysis to optimize motorcycle rear-geometry and sizing for improved performance.

Cal Aero SAE | Wing Design Engineer Sep 2023 – Apr 2024

- Worked with Berkeley students to design, build, test, and fly a RC plane for the Aero SAE competition in regular class.
- Helped develop an aerodynamically efficient wing model through Solidworks, enhancing flight performance.
- Researched aerodynamic characteristics such as dihedral and wing position for performance through Matlab scripts.
- Conducted bolt shear analysis for Spar integration points.
- Helped laser cut airfoils and manufacture wing structure used in flown model.

Glendale Community College | Physics Laboratory Technician Oct 2022 – Dec 2022

- Tested and maintained equipment for physics labs such as oscilloscopes, multimeters, sensors, and circuit components to ensure proper functionality for safety and reliable results.
- Reviewed and suggested improvements to the existing labs if necessary.
- Assisted instructor and students with any and all lab processes, questions, or concerns for optimal lab procedures.

BV Fire Design | Fire Protection Engineer Aug 2020 – Aug 2021

- Learned HydraCAD to assist in analyzing architectural and mechanical plans for sprinkler systems.
- Aided in job site surveys.
- Spearheaded residential design prep and communicated with water and fire department officials.
- Organized administrative paperwork and applications.

PROJECTS

πRo-Bot: Autonomous Fire Suppression Robot | *(Fusion 360, Motor Control, 3D Printing, Circuit, Soldering, Arduino IDE)*

- Designed and developed πRo-Bot, an autonomous fire suppression system featuring infrared sensing, real-time positioning, and remote operation; led the electrical system design encompassing circuitry, power distribution, and microcontroller integration to enable seamless sensor–actuator communication.
- Directed manufacturing, assembly, and system integration, ensuring reliable mechanical–electrical interfacing and improving fire suppression accuracy through closed-loop feedback control.

Performance Assessment of Thermal Paste | *(MATLAB, ESP32 Microcontrollers, Calibration, Heat Transfer, Circuit)*

- Designed and executed a controlled experiment to assess thermal paste conductivity using custom aluminum blocks, K-type thermocouples, and an ESP32-based data acquisition system.
- Analyzed temperature gradients, demonstrating that the most effective thermal paste reduced thermal resistance by 79% compared to no-paste setups and by 51% relative to other industry-standard pastes.

Microfluidic Exhaust Valve | *(AutoCAD, MEMS, Circuit)*

- Designed a MEMS-based microfluidic valve incorporating a yoke array of six electrothermal actuators for precise fluid control, reducing leakage in microscale channels through an electrostatic latching mechanism.

Efficient Wind Turbine | *(3D Printing, SOLIDWORKS, Mechanical Drawings, Finite Element Analysis (FEA))*

- Designed, fabricated, and tested a small-scale wind turbine focused on maximizing power output and stiffness while minimizing weight; achieved a peak power output of 1.56 W under 25 mph wind conditions.
- Engineered a 3-blade rotor using the SG6043 airfoil with a 13° angle of attack and 12° twist angle, and developed a lightweight 3D-printed ABS tower featuring a cross-braced truss design that exhibited 18.51 N/mm stiffness and only 0.53 mm deflection under a 1 kg load.
- Utilized SolidWorks for CAD modeling and FEA simulations to validate structural performance, integrating experimental testing and data acquisition for design verification and identifying improvements in truss geometry and wind capture efficiency.